

Tahsin R. Chowdhury

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EDUCATION

The University of Texas at Arlington

Arlington, TX

Bachelor of Science in Information Systems, STEM-Designated, AACSB-Accredited

Aug. 2019 – May 2024

Relevant Coursework: Full-Stack Development, Database Systems, Data Structures & Algorithms, Software Engineering SDLC, Systems Analysis & Design, Information Security, Cloud Computing

Awards: Maverick Academic Scholarship, Honor Roll, Dean's List

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java (OOP, SOLID, JDBC, JUnit), SQL, HTML5, CSS3

Frameworks & Web: React, Next.js, Node.js, Express.js, REST APIs, HTTP, front-end/back-end architecture

Tools & Platforms: PostgreSQL, Git, GitHub (branching, pull requests, code review), Jest, AWS (S3, Lambda, CloudFront), Docker, Agile/Scrum, Jira

EXPERIENCE

Self-Employed

Jan. 2024 – Present

Software Engineer

Arlington, TX

- Wrote clean, testable, maintainable code across 5 production applications (JavaScript, TypeScript, Python, Node.js, PostgreSQL, AWS) — Jest unit tests, Zod schema validation, GitHub pull request code review, and CI/CD on every project. Debugged a production frame-rate issue via Chrome DevTools, identified root cause across 300+ concurrent data points, and eliminated drops with a Web Worker fix.

University of Texas at Arlington

Jan. 2022 – May 2024

Undergraduate Research Assistant / Software Developer

Arlington, TX

- Gathered requirements from research stakeholders, translated them into a normalized SQL database tracking 1,000+ employee records, and built Python automation workflows that cut manual reporting time by 30% — queried the database with joins and aggregations to validate data integrity and catch discrepancies before they reached stakeholders.
- Collaborated across 4 Agile software projects in JavaScript, TypeScript, React, Node.js, Java, and SQL — writing unit and integration tests, submitting pull requests, participating in sprint planning and retrospectives, and maintaining technical documentation throughout delivery.

PROJECTS

FUN – Real-Time Sports Platform | [Live Demo](#) | [GitHub](#) | *JavaScript, TypeScript, Node.js, Express.js, PostgreSQL, REST*

- Features from requirements to deployment — PostgreSQL schema, REST API, real-time messaging, geo venue discovery. Diagnosed production frame-rate issue via Chrome DevTools, fixed with Web Worker, wrote unit tests covering the solution.

Summarist – SaaS Platform | [Live Demo](#) | [GitHub](#) | *TypeScript, React, Next.js, Node.js, Firebase Auth, Stripe, REST APIs*

- RBAC gated from auth token through SSR boundaries, Stripe billing with subscription lifecycle, real-time data sync, data integrity at every layer. Architecture decisions documented throughout.

Library Management System | [GitHub](#) | *PostgreSQL, SQL, Stored Procedures, CTEs, Window Functions*

- Defined requirements, modeled a 6-table normalized schema with acceptance criteria per workflow, built end-to-end in PostgreSQL — 20 analytical queries using CTEs and window functions validated business logic before sign-off.

Nike E-Commerce | [Live Demo](#) | [GitHub](#) | *TypeScript, Next.js, PostgreSQL, Drizzle ORM, Stripe, Zod*

- Zod validation on all API payloads, JWT-secured OAuth 2.0, SSR/CSR boundaries, SQL schema design with Drizzle ORM, CI/CD deployment.

LEADERSHIP & ACTIVITIES

UT Arlington Student Government

Jan. 2022 – May 2023

Senator, College of Business

Delta Sigma Phi

Jan. 2021 – Dec. 2023

Chancellor